

Ruoyu Wen

PhD Candidate in Product Design — Human-Computer Interaction

Human-Agent Interaction · Game Design · Virtual Reality · Generative AI

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[Google Scholar](#) · [LinkedIn](#) · [Portfolio](#)

PROFILE

HCI researcher and game designer working at the intersection of **human-agent interaction**, **virtual reality**, **alternate reality games (ARGs)**, and **generative AI**. I combine rigorous mixed-methods research with hands-on game development to design playful, intelligent systems that foster meaningful connection and cross-cultural understanding. Author of **15+ peer-reviewed publications** at venues including CHI, IUI, CHI Play, DiGRA, OzCHI, and SIGGRAPH Asia, with international research experience across New Zealand, Germany, Korea, Sweden, and China.

Research Interests: Human-Agent Interaction · Virtual & Extended Reality · Alternate Reality Games · Player Psychology (Self-Determination Theory) · Conversational & Generative AI · Cross-cultural HCI

Languages: Chinese (Native) · English (Fluent)

EDUCATION

University of Canterbury — Christchurch, New Zealand

PhD in Product Design, HIT Lab NZ · 2023 – 2027 (expected Nov 2027)

Human-agent interaction, VR, and player motivation in ARGs through Self-Determination Theory; mixed-methods experimental research.

Advisors: [Assoc. Prof. Thammathip Piumsomboon](#), [Assoc. Prof. Simon Hoermann](#), [Prof. Mark Billinghurst](#).

Uppsala University — Uppsala, Sweden

Master of Arts in Game Design · 2021 – 2023

Interactive narrative, player psychology, user experience, and game design theory.

Advisor: [Assoc. Prof. Sarah Lynne Bowen](#).

Huazhong University of Science and Technology — Wuhan, China

Bachelor in Communication Studies · 2016 – 2021

Digital media, social media, human-computer interaction, and digital culture.

Advisor: [Assoc. Prof. Shuo Xiong \(熊硕\)](#).

HONORS & GRANTS

Research Grants in Germany — DAAD (Deutscher Akademischer Austauschdienst) · 2026

Competitive research grant supporting a research stay at the University of Augsburg, Chair for Human-Centered AI.

Chinese Government Award for Outstanding Self-Financed Students Abroad — China

Scholarship Council (CSC) · 2025

National award recognising outstanding self-funded PhD students overseas (University of Canterbury, New Zealand).

RESEARCH & PROFESSIONAL EXPERIENCE

Visiting PhD Student — University of Augsburg, Chair for Human-Centered AI (Prof. Elisabeth André), Augsburg, Germany

Apr 2026 – Present

- Collaborating on human-centered AI, intelligent user interfaces, and human-agent interaction.

Lead AI Game Designer — MICROFEEL, Remote

Oct 2024 – Present

- Led a cross-functional team building an end-to-end system that generates game stories, mechanics, and NPCs from natural-language input.
- Translated generative-AI research into shipping product features.

Visiting PhD Student — KAIST, GSCT UVR Lab, Daejeon, South Korea

Oct 2025

- Collaborative research on extended reality (XR) and human-computer interaction.

Teaching Assistant — University of Canterbury, Christchurch, New Zealand

Jul 2025 – Oct 2025

- TA for *Computation for Games*; mentored undergraduates in game development, programming, and computational foundations.

PhD Researcher / Research Assistant — University of Canterbury, HIT Lab NZ, Christchurch, New Zealand

2023 – Present

- Designed and ran VR and human-agent interaction studies on conversational AI, including experiment design, data collection, and mixed-methods analysis.
- First-author and co-author publications at CHI, OzCHI, DiGRA, and SIGGRAPH Asia.

Research Intern — NeXT SCENE (Immersive Entertainment), Shanghai, China

Apr 2022 – May 2022

- Studied narrative-driven entertainment formats (script murder, escape rooms) and player engagement in social gaming.

Strategy Investor Intern — Huya (NASDAQ: HUYA), Guangzhou, China

Feb 2021 – Jun 2021

- Market research and investment analysis across gaming and entertainment technology.

Game Designer — Chengdu ChuangLeHui (JOY NOVA), Sichuan, China

Mar 2020 – May 2020

- Concept design, gameplay design, and playtesting for the mobile game *Champion of the Fields*.

SELECTED PUBLICATIONS

For the complete list, see [Google Scholar](#).

1. **AI of Oz: Enhancing Wizard of Oz Studies in HCI with AI Assistance for Human Moderation.** *ACM CHI*, 2026.
2. **From Prompt to Presence: Co-Creating Personalised Emotional Sanctuaries in VR with Generative AI.** *ACM IUI*, 2026.
3. **Can AI Prompt Humans? Multimodal Agents Prompt Players' Game Actions and Show Consequences to Raise Sustainability Awareness.** *ACM CHI*, 2025.
4. **Exploring User Preferences for Museum Guides: The Role of Chatbots in Shaping Interactive Experiences.** *CHI Extended Abstracts*, 2025.
5. **GenLinguaScape: Enabling User-Defined VR Scenarios for Communicative Language Practice.** *IEEE ISMAR Adjunct*, 2025.
6. **Large Language Models for Automatic Detection of Sensitive Topics.** *OzCHI*, 2024.
7. **An Exploratory Study on AI-driven Visualisation Techniques for Decision Making in Extended Reality.** *OzCHI*, 2024.
8. **Sketchar: Supporting Character Design and Illustration Prototyping Using Generative AI.** *Proc. ACM HCI (CHI Play)*, 2024.

9. **Motivational Landscapes of ARG Players: A Self-Determination Theory Perspective.** *DiGRA*, 2024.
10. **Virtual Triplets: Mixed-Modal Synchronous and Asynchronous Collaboration with Human-Agent Interaction in VR.** *CHI Extended Abstracts*, 2024.
11. **I Light U Up: Exploring a New Emergent Narrative Paradigm through Light and Generative AI.** *SIGGRAPH Asia Art Papers*, 2024.
12. **Listen to the Sword: Using Breathing and Spatial Audio for Wuxia Games.** *CHI Extended Abstracts*, 2024.
13. **The Impact of Alternate Reality Games on Environmental Cognition for University Freshmen.** *HCI International*, 2024.
14. **Player Category Research on Murder Mystery Games.** *International Journal of Role-Playing*, 2023.
15. **The Chinese Hotpot of Larp.** *Knutpunkt Magazine*, 2022.